

E.g. 1.

Prove that if 101 integers are selected from the set $S = \{1, 2, \dots, 200\}$, then two integers are such that one divides the other.

For $x \in S$ we write it as $x = 2^k y$,

s.t. $k \geq 0$ & $\gcd(2, y) = 1$.

$y = \{1, 3, 5, \dots, 199\}$

$|y| = 100$.

101
y y

Since we have choose 101 no.s from S these will be at least two numbers

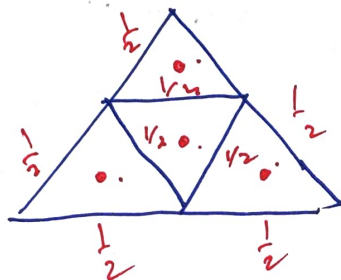
$a = 2^n y$, $b = 2^m y$ for which
y should repeat (by PHP).

if $n > m$ then $b|a$ &
if $m > n$ then $a|b$. $\square$

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E.g. 2

Triangle ACE is equilateral with $AC=1$.
If five points are selected from the interior,
there are at least two whose distance
apart is less than $\frac{1}{2}$.



E.g. 3.

Find a subarray whose sum is divisible by the size of the array (N) itself.

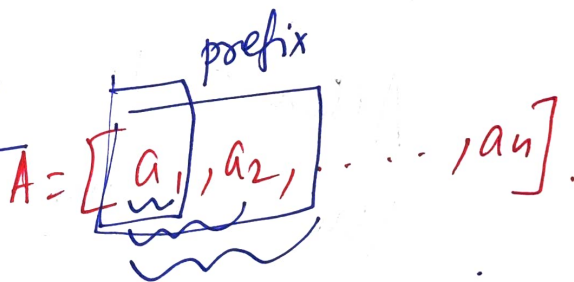
Brute force approach:

Find all subarrays $\rightarrow$ sum $\rightarrow$
flag the start & end indices.

$\rightarrow O(N^3)$.

Better approach:

Use prefix sum.



$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$\vdots$$
$$S_n = a_1 + a_2 + \dots + a_n$$

$$S[i, j] = \text{prefixsum}[j] - \text{prefix}[i-1].$$

$\rightarrow O(N^2)$.

Can we do even better?

Consider the remainders of dividing each prefix sum by N .

If none results in remainder 0, then the only possible remainders

are $1 - (N-1)$

N prefix sums, $(N-1)$ remainders.

At least two prefix sums will have the same remainder.

Subtract these $\rightarrow$ A subarray sum that is perfectly divisible by N . $O(N)$. $\rightarrow$ H.W. write the code & execute it.

$\rightarrow$ Such a subarray shall always exist.

Defⁿ Generalized PHP.

If N objects are placed in K boxes, then
at least one box will contain ^{at least} $\lceil N/K \rceil$ objects
if $N > K$.

Proof: By way of contradiction.

Assume, if not, each box must have less than
 $\lceil N/K \rceil$ objects $\rightarrow$ at most $(\lceil N/K \rceil - 1)$ objects

Since there are K boxes $\rightarrow \lceil N/K \rceil - 1$ objects
in each box $\rightarrow$ Total no. of objects

~~$K \cdot \lceil N/K \rceil = N$~~

$K \left(\lceil N/K \rceil - 1 \right) < K \cdot \frac{N}{K} = N.$

$$\lceil x \rceil - 1 < x.$$

Contradiction.

□.

E.g 5.

If 5 colors are used to paint 26 doors, show that at least 6 doors have same color.

$$n = 26, \quad k = 5$$

$$\left\lceil \frac{n}{k} \right\rceil = \left\lceil \frac{26}{5} \right\rceil = \left\lceil 5.2 \right\rceil = 6.$$

H.W. Let $m \in \mathbb{Z}^+$ with m odd. Prove that there exists a positive integer n such that m divides $2^n - 1$.

→ Hint: use PHP + division algo.

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set.

Set: Collection of unique elements.

Representation.

Roster form: List out all the elements of the set
 $O = \{1, 3, 5, 7, 9, \dots\}$.

Set builder notation: Tries to establish a relation among the elements of the set
 $E = \{y \mid y = 2x, x \in \mathbb{N}\}$.

Subsets & Universe.

$U \rightarrow$ set of all possible elements/items.

$C \subseteq D$, $D \supseteq C \Rightarrow$ every element in C is also in D .

$C \subset D \rightarrow$ there are some elements in D that are not in C
 $\hookrightarrow$ proper subset.

Two sets C & D are identical iff
 $C \subseteq D$ & $D \subseteq C$.

$$\underbrace{\hspace{10em}}_{C=D.}$$

Theorem 1.

If $A, B, C \subseteq U$

- (a) $A \subseteq B$ & $B \subseteq C \Rightarrow A \subseteq C$.
- (b) $A \subset B$ and $B \subseteq C \Rightarrow A \subset C$
- (c) $A \subseteq B$ & $B \subset C \Rightarrow A \subset C$
- (d) $A \subset B$ and $B \subset C \Rightarrow A \subset C$.

Read:

$$P \subseteq Q.$$

for all $x \in U$

if $x \in P$ then $x \in Q$.

(a) $A \subseteq B \Rightarrow \left. \begin{array}{l} \text{if } x \in A \xrightarrow{\text{then}} x \in B \\ \text{if } x \in B \xrightarrow{\text{then}} x \in C \end{array} \right\}$
 $B \subseteq C \Rightarrow$
 $x \in A \rightarrow x \in C$. Law of ~~Syllogism~~
 $\therefore A \subseteq C$. D. Syllogism.

(b) $B \subseteq C : \forall x, x \in B \Rightarrow x \in C$.
 $A \subseteq B : \frac{\exists x, x \in B \wedge x \notin A}{\exists x, x \in C \wedge x \notin A}$
 $\therefore A \subseteq C$. D.

Defⁿ Null set / Empty set.
 Null set / Empty is a set containing no elements.
 Φ or $\{\}$.
 $|\Phi| = 0$ $\{0\} \neq \Phi$ $\Phi \neq \{\Phi\}$
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Theorem:

For any universe U , let $A \subseteq U$.
Then $\bar{\Phi} \subseteq A$ and if $A \neq \bar{\Phi}$ then
 $\bar{\Phi} \subset A$.

Proof: By way of contradiction.

$$\bar{\Phi} \not\subseteq A.$$

There is an element $x \in U$ ~~such~~ such
that $x \in \bar{\Phi}$ and $x \notin A$.

But $x \in \bar{\Phi}$ is impossible.

We reject the hypothesis &
 $\bar{\Phi} \subseteq A$ $\square$.

Further if $A \neq \bar{\Phi}$ then these must
be some element $a \in A$ & $a \notin \bar{\Phi}$

$$\Rightarrow \bar{\Phi} \subset A. \quad \square$$

Defⁿ Power set.

If A is a set of elements from U , then the power set of A , $P(A)$ is the collect of all subsets of A .

For a finite A , $|A| = n \geq 0$

$$|P(A)| = 2^n$$

$$\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n \text{ for } n \geq 0.$$

Defⁿ Set operations.

- (a) $A \cup B : \{x \mid x \in A \vee x \in B\}$
- (b) $A \cap B : \{x \mid x \in A \wedge x \in B\}$
- (c) $A \Delta B = \text{Symmetric Difference.}$
 $\{x \mid x \in A \cup B \wedge x \notin A \cap B\}$

Defⁿ

Disjoint sets.

$S, T \subseteq U$ The sets S & T are called disjoint / mutually exclusive when $S \cap T = \emptyset$

Theorem: If $S, T \subseteq U$ then S & T are disjoint iff $S \cup T = S \Delta T$.

Forward $\Rightarrow$

$$S \cap T = \emptyset$$

If $x \in S \cup T$ & given $x \notin S \cap T$
we have $x \in S \Delta T$.

$$S \cup T \subseteq S \Delta T. \text{ — (1)}$$

If $y \in S \Delta T$, $y \notin S \cap T (\underline{\emptyset})$

$y \in S$ or $y \in T$.

$\therefore y \in S \cup T$.

$$\therefore S \Delta T \subseteq S \cup T. \text{ — (2)}$$

$\therefore$ From (1) & (2) $S \Delta T = S \cup T$.

← Backward.

Given $S \cup T = S \Delta T.$

By way of contradiction: $S \cap T \neq \underline{\emptyset}$

$\therefore x \in S \cap T.$

also $x \in S \cup T.$

$\therefore x \in S \Delta T$ ($\neq S \cup T$)

But if $x \in S \cup T$ & $x \in S \cap T$
 $x \notin S \Delta T.$

Both cannot be true $\begin{cases} x \in S \Delta T \\ x \notin S \Delta T \end{cases}$

$\therefore S \cap T = \underline{\emptyset}$

Definition:

Complement of a set.

$A \subseteq U$, then complement of A , $\bar{A}$

or $U - A$ is given by

$\{x | x \in U \wedge x \notin A\}.$

Defn

Relative Complement.

$A, B \subseteq U$, then the relative complement
or $B - A$ is given by

$$\{x \mid x \in B \wedge x \notin A\}.$$

Theorem:

$$(A \subseteq B) \Rightarrow (A \cup B = B) \Rightarrow (A \cap B = A) \\ \Rightarrow (\overline{B} \subseteq \overline{A}).$$

Let us know $B \subseteq A \cup B$.

$$x \in A \cup B \Rightarrow x \in A \text{ or } x \in B.$$

$$A \subseteq B.$$

Any ~~any~~ $x \in B$.

$$A \cup B \subseteq B.$$

$$\therefore A \cup B = B.$$

Laws of set theory.

- ① $\overline{\overline{A}} = A$
- ② $\overline{A \cup B} = \overline{A} \cap \overline{B}$
 $\overline{A \cap B} = \overline{A} \cup \overline{B}$
- ③ $A \cup B = B \cup A$
 $A \cap B = B \cap A$
- ④ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- ⑤ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 ~~$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$~~
- ⑥ $A \cup A = A$
 $A \cap A = A$
- ⑦ $A \cup \Phi = A$
 $A \cap U = A$
- ⑧ $A \cup \overline{A} = U$
 $A \cap \overline{A} = \Phi$
- ⑨ $A \cup U = U$
 $A \cap \Phi = \Phi$
- ⑩ $A \cup (A \cap B) = A$
 $A \cap (A \cup B) = A$

Defⁿ ~~Duality~~ Duality of sets.

Let s be general statement dealing with some equality of set expressions.

Each such expression may involve the occurrence of one or more sets

(e.g., $A, \bar{A}, B, \bar{B}, \dots$) and one or more occurrence of $\emptyset$ or U .
Also the only set operations allowed are $\cup$ & $\cap$

The dual of s , s^d is obtained by replacing each

- ① $\cup$ by $\cap$
- ② $\cap$ by $\cup$
- ③ $\emptyset$ by U
- ④ U by $\emptyset$

Principle of duality: If s is a theorem involving equality of set expressions, so is s^d .

Defⁿ

Index sets.

Let I be a non-empty ~~subset of~~ U . For each element $i \in I$ let $A_i \subseteq U$.

Then I is called the index set (set of indices). Each $i \in I$ is called an

INDEX.

$$\bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for at least one } i \in I\}.$$

$$\bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for every } i \in I\}.$$

If the index set $I \in \mathbb{Z}^+$

$$\bigcup_{i \in \mathbb{Z}^+} A_i = A_1 \cup A_2 \dots = \bigcup_{i=1}^{\infty} A_i.$$

$$\bigcap_{i \in \mathbb{Z}^+} A_i = A_1 \cap A_2 \dots = \bigcap_{i=1}^{\infty} A_i.$$

Theorem: Generalized De Morgan's Law.

$$(a) \overline{\bigcup_{i \in I} A_i} = \bigcap_{i \in I} \overline{A_i}$$

$$(b) \overline{\bigcap_{i \in I} A_i} = \bigcup_{i \in I} \overline{A_i}$$

Proof: For each $x \in U$

$$x \in \overline{\bigcup_{i \in I} A_i} \Leftrightarrow x \notin \bigcup_{i \in I} A_i \Leftrightarrow x \notin A_i$$

for all i

For all $i \in I$, $x \in \overline{A_i}$
 $\Rightarrow$ all $i \in I$, $x \in \bigcap_{i \in I} \overline{A_i}$

Defⁿ:

Set counting:

If A & B are finite sets then:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

if A & B are disjoint $\Rightarrow |A \cup B| = |A| + |B|$.

$$|\overline{A \cap B \cap C}| = |\overline{A \cup B \cup C}|$$

$$= |U| - |A \cup B \cup C|$$

$$= |U| - |A| - |B| - |C| + |A \cap B| + |A \cap C| + |B \cap C| - |A \cap B \cap C|$$

E.g. A student visits an arcade each day after school (5 day school) & plays one ~~game~~ game of either Laser Man, Millipede, Space Conquerers. In how many ways can she play one game each day so she plays each of the three types at least once during the school week?

$U \rightarrow$ all arrangements of size five taken from the set of 3 games.

$A \rightarrow$ subset of all sequences of five games played during the week with out playing Laser man.

$B \rightarrow \dots$ leaving out Millipede

$C \rightarrow \dots$ leaving out SC.

$$|U| = 3^5$$

$$|A| = 2^5$$

$$|B| = 2^5$$

$$|C| = 2^5$$

$$|A \cap B| = 1^5$$

$$|B \cap C| = 1^5$$

$$|A \cap C| = 1^5$$

$$|A \cap B \cap C| = 0$$

$$|\overline{A} \cap \overline{B} \cap \overline{C}| = 3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 - 0 = 150$$

Ans.